

Camera Shots & VFX – Exam Night Summary ★

1. VFX (Visual Effects) ★★★★★

Definition:

VFX is the integration of live-action footage with computer-generated imagery (CGI) to create scenes that are difficult, expensive, dangerous, or impossible to film.

Uses of VFX

- Add impressive content to movies.
- Create imaginary environments.
- Add special effects to movies.

2. Difference Between VFX and 3D Animation ★★★★★

VFX	3D Animation
Combines real footage + CGI	100% CGI
Uses live-action plates	Fully computer-generated
Example: The Avengers	Example: Toy Story

3. Difference Between VFX and SFX ★★★★★

VFX	SFX (Special Effects)
Requires computers	Physical effects on set
Done after shooting	Done during shooting
Digital effects	Practical effects
Example: Dragons, Spaceships	Explosions, Fake wounds

Common MCQ:

VFX = Digital

SFX = Practical

4. Main Types of VFX ★★★★★

1. CGI (Computer Generated Imagery)

- Most common VFX type.
- Creates objects/scenes using 3D models.
- Actions created by computer without performing them in reality.

2. Compositing & Green Screen

- Combines multiple images into one image.
- Also called Chroma Keying.
- Requires green or blue screen.

3. Motion Capture (Mocap)

- Combines live action with CGI.
- Captures actor movements and transfers them to digital characters.

5. Motion Capture (Very Important) ★★★★★

Motion Detection Methods

- Computer Vision
- Special Markers
- Sensors

Components of Motion Capture

1. Mocap Suit

- Captures body movement.
- Contains sensors (15–20).
- Sends movement data to software in real time.

2. Head Mounted Camera

- Captures facial expressions.
- Can be Optical or Infrared.

3. Software

- **Examples:**
 - Autodesk MotionBuilder

- 3DS Max

6. Compositing ★★★★★

Definition

Combining two or more images so they appear as a single image.

Example:

Weather presenter in front of green screen.

7. Important Compositing Techniques ★★★★★

- **Rotoscope Animation**
 - Tracing live-action footage frame by frame.
- **Chroma Keying**
 - Removing a specific color (usually green).
 - Replacing it with another background.
- **Double Exposure**
 - Combines two images together.
 - Creates ghost-like effect.
- **Bullet Time**
 - Creates dramatic slow-motion effect.
- **Doubling**
 - Creates twins or clones from one actor.

8. Natron Software ★★★★★

What is Natron?

- Open-source node-based compositor.
- Uses nodes connected by lines.

Node Rules

- Node can have one or more inputs.
- Many nodes have a mask input.
- Every node has exactly one output.
- Viewer node has no output.

MCQ Favorite

Which node has no output?

➡ Viewer Node

9. Merge Node ★★★★★

Used to combine images.

Steps

1. Add Merge Node.
2. Connect image A and B.
3. Connect Viewer.
4. Select merge operation.

Most Common Operation

Over

Places image A over image B.

10. Important Merge Operations ★★★★★

Operation	Function
Over	A over B
Atop	A covers overlap area
Average	Average of two images
Copy	Shows only A
Difference	Shows pixel differences
From	$B - A$
In	Intersection only
Mask	Reverse of In
Max	Maximum values
Min	Minimum values
Color Burn	Darkens image
XOR	Shows non-overlapping areas

Very Common MCQs

- Over = A over B
- Copy = Only A
- Difference = Compare images
- In = Intersection
- Mask = Reverse of In
- XOR = Non-overlapping regions

11. Rendering Output ★★★★★

To export final video:

Required Node

Write Node

Then click:

Render

to save the final file.

12. Software Names to Memorize ★★★★★

VFX Software

- Adobe After Effects
- Autodesk Maya
- 3Ds Max
- Houdini
- Cinema 4D
- Nuke
- Natron

Open Source VFX Software

- Blender
- Natron
- Fusion

- Gaffer

Last-Minute Questions Expected in Exam

- Define VFX.
- Difference between VFX and SFX.
- Difference between VFX and 3D Animation.
- List VFX Types.
- What is CGI?
- What is Compositing?
- What is Chroma Keying?
- What is Motion Capture?
- Components of Motion Capture.
- What is Rotoscope Animation?
- What is Double Exposure?
- What is Bullet Time?
- What is Doubling?
- What is Natron?
- Node Rules.
- What is Merge Node?
- Explain Over, Difference, In, Mask, XOR operations.
- What node is required before rendering? → Write Node.